

Prediction of Strouhal Number for Flow Over an Elliptic Cylinder Using CFD and Neural Networks

Aditya Prasad (22110018)

Department of Mechanical Engineering, IIT Gandhinagar

Instructor: Prof. Vinod Narayan

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1. Abstract

This project explores the aerodynamic behavior of flow over an elliptic cylinder in Water. Computational simulations were conducted by varying the angle of attack, Reynolds number, and aspect ratio of the cylinder. The Strouhal number corresponding to vortex shedding was calculated for each case. A simple multilayer perceptron (MLP) neural network was trained to predict the Strouhal number based on three input parameters: aspect ratio, Reynolds number, and angle of attack.

2. Introduction

Flow past bluff bodies like elliptic cylinders exhibits vortex shedding characterized by the Strouhal number. Predicting this number accurately helps in understanding aerodynamic instabilities and resonance. Traditional methods rely on experiments or high-fidelity CFD simulations. This project combines parametric CFD simulations with machine learning to build a predictive model for the Strouhal number.

3. Simulation Setup

We explain the simulation setup for the flow past a elliptic cylinder. The aspect ratio (i.e. length of major axis / minor axis) of the elliptic cylinder is set to 2. We try to set the initial flow velocity and chose the flow medium in such a way to ensure the flow is in the turbulent regime and there is vortex shedding. In this case, we try to get a turbulent flow with a Reynolds number of 100.

3.1. Geometry and Domain

The computational domain was chosen to be $12\text{ m} \times 8\text{ m}$, with an embedded elliptic cylinder of major axis 1 m and minor axis 0.5 m . The ellipse was oriented according to the specified angle of attack.

We chose a comparatively large fluid as compared to the size of the elliptic cylinder. We did this because we would need sufficiently large flow domains to capture all of the flow conditions, like vortex shedding.

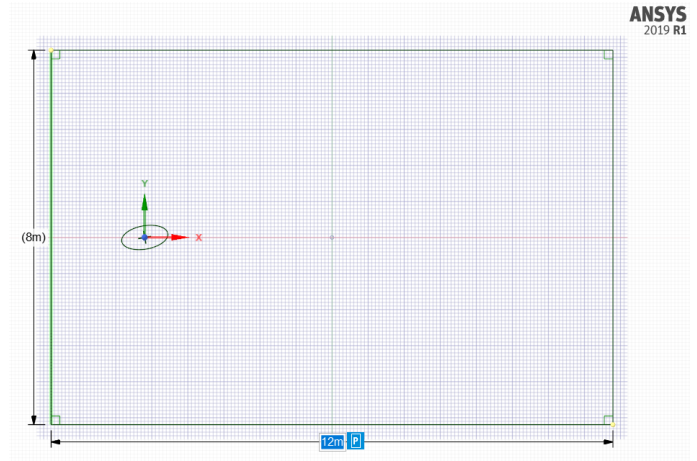


Figure 1. Flow Domain

3.2. Meshing

Meshing was performed in ANSYS Mesher with special attention to resolving the boundary layer and wake region accurately.

Element Size Selection

First, the element size on the entire face was set to 0.1 m . To better capture flow details near the cylinder surface, the element size on the `cylinder_wall` boundary edge was refined to 0.025 m .

Inflation layers

An inflation layer mesh was then added to the `cylinder_wall` boundary to capture near-wall gradients accurately. The inflation parameters were:

- Number of layers: 20
- Growth rate: 1.5
- First layer thickness: 0.025 m

Naming Edges

All boundaries were explicitly named:

- `inlet`
- `outlet`
- `walls`
- `cylinder_wall`

After these steps, the mesh was generated and saved for use in Fluent. The final mesh offered a good balance between accuracy and computational cost.

The attached image below shows the final generated mesh:

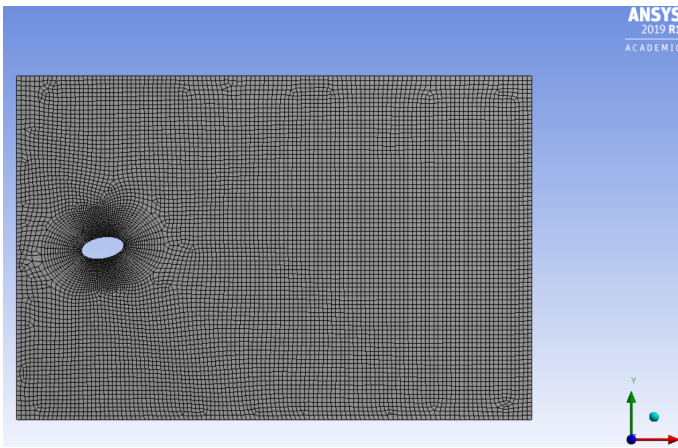


Figure 2. Mesh domain

3.3. Solver and Model Selection

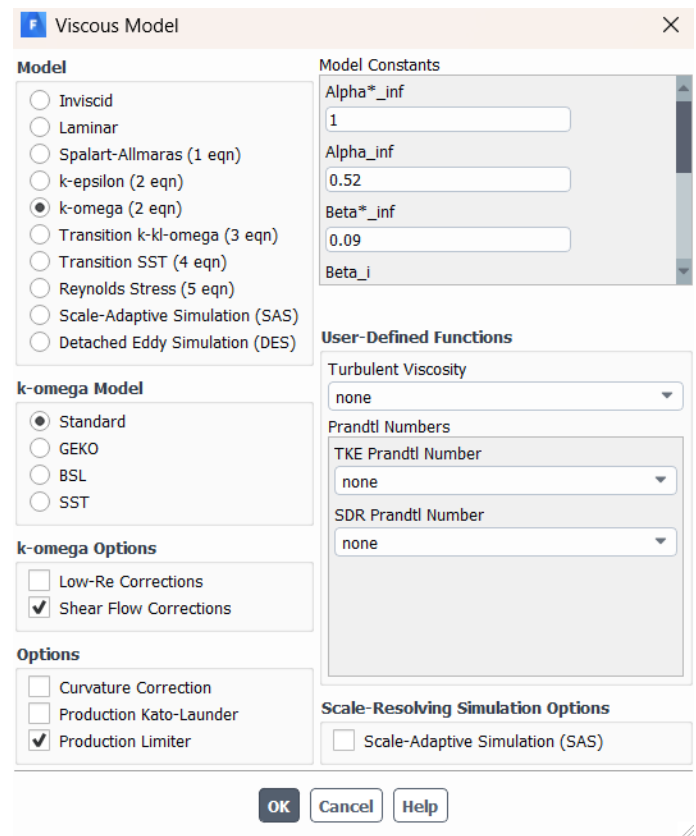
The solver and physical models were carefully selected to accurately simulate the transient vortex shedding behind an elliptic cylinder at low Reynolds numbers. The following subsections explain each choice in detail.

Pressure-Based Transient Solver

A pressure-based transient solver was used because the flow was incompressible and involved time-dependent vortex shedding. The transient formulation allows capturing unsteady phenomena such as alternating vortex formation and lift coefficient oscillations, which are essential for Strouhal number evaluation.

Turbulence Model: $k-\omega$ (2-Equation)

The viscous turbulence model was set to $k-\omega$ (2eqn), particularly the SST (Shear Stress Transport) variant. This model is well-suited for low Reynolds number flows involving boundary layer separation and adverse pressure gradients. It provides better accuracy than the standard $k-\epsilon$ model for external, 2D, and low-speed transient flows.


 Figure 3. $k-\omega$ model

Material Selection

The working fluid chosen was **Water (liquid)**, selected from the ANSYS Fluent materials database.

Water was selected due to the following reasons:

- It is a commonly studied fluid in low-speed hydrodynamic simulations.
- Its physical properties (density and viscosity) are well-documented and stable over a wide temperature range.
- The relatively high viscosity of water at room temperature allows for simulating low Reynolds number flows with practical inlet velocities.

To verify that the simulation is within the laminar or low-turbulence regime, the Reynolds number was computed using the formula:

$$Re = \frac{\rho U D}{\mu}$$

where:

- $\rho = 998.2 \text{ kg/m}^3$ (density of water at 20°C)
- $\mu = 0.001003 \text{ Pa s}$ (dynamic viscosity of water at 20°C)
- $U = 0.0001 \text{ m s}^{-1}$ (inlet velocity)
- $D = 1.0 \text{ m}$ (characteristic length = major axis of ellipse)

Substituting the values:

$$Re = \frac{998.2 \times 0.0001 \times 1.0}{0.001003} \approx 99.5$$

Thus, the Reynolds number is approximately 100, confirming that the chosen velocity and fluid properties lead to the intended flow regime. This validates the use of water as a suitable working fluid for simulating low-Reynolds-number vortex shedding.

Boundary Conditions

Boundary conditions were defined as follows:

- **Inlet:** A uniform absolute velocity of 0.0001 m s^{-1} was applied to ensure a Reynolds number of approximately 100 based on the major axis length (1 m).
- **Cylinder Wall:** No-slip wall condition to accurately simulate boundary layer effects.
- **Outlet:** Pressure outlet with default static gauge pressure.
- **Walls:** No slip shear condition through the stationary wall.

Time Step and CFL Number Justification

The simulation was performed using a time step of 100 s, with a total of 6000 time steps to ensure sufficient flow development and vortex shedding cycles.

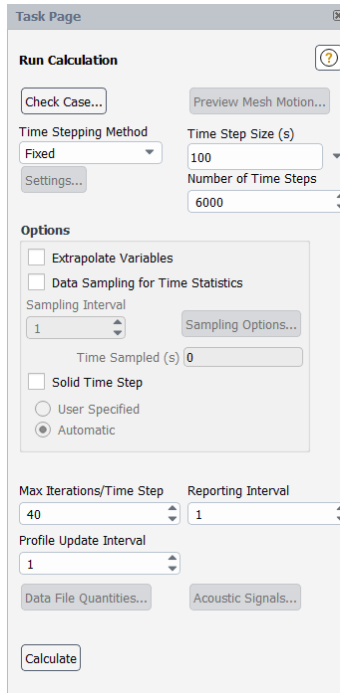


Figure 4. Time step selection

The time step was selected based on the Courant–Friedrichs–Lewy (CFL) condition:

$$\text{CFL} = \frac{U \cdot \Delta t}{\Delta x}$$

where:

- $U = 0.0001 \text{ m s}^{-1}$
- $\Delta t = 100 \text{ s}$
- $\Delta x = 0.025 \text{ m}$

Substituting the values:

$$\text{CFL} = \frac{0.0001 \times 100}{0.025} = 0.4$$

A CFL number less than 1 is considered stable for explicit schemes and adequately conservative for capturing transient behavior. Thus, the selected time step of 100 s ensures numerical stability while allowing the flow features to evolve smoothly.

Output Controls and Data Collection

Report files were configured to extract time histories of drag coefficient (C_d) and lift coefficient (C_l) on the cylinder wall. These time-dependent signals were later used to calculate vortex shedding frequency via FFT and compute the Strouhal number.

Run the Calculation..

Start the calculation and ensure that the laptop sleep time is more than 3 hours. After 2-3 hours of continuous computation we can move to the data collection phase.

4. Post Processing

Post-processing was carried out to visualize the unsteady behavior of the flow over the elliptic cylinder. Contour plots of velocity magnitude and pressure were generated at selected time steps to observe the formation and evolution of vortex structures in the wake.

Velocity Contours at Different Time Steps

Figure 5 shows the development of vortex shedding through velocity magnitude contours at different time instances: $t = 0$, 1000, 4000, and 6000 seconds.

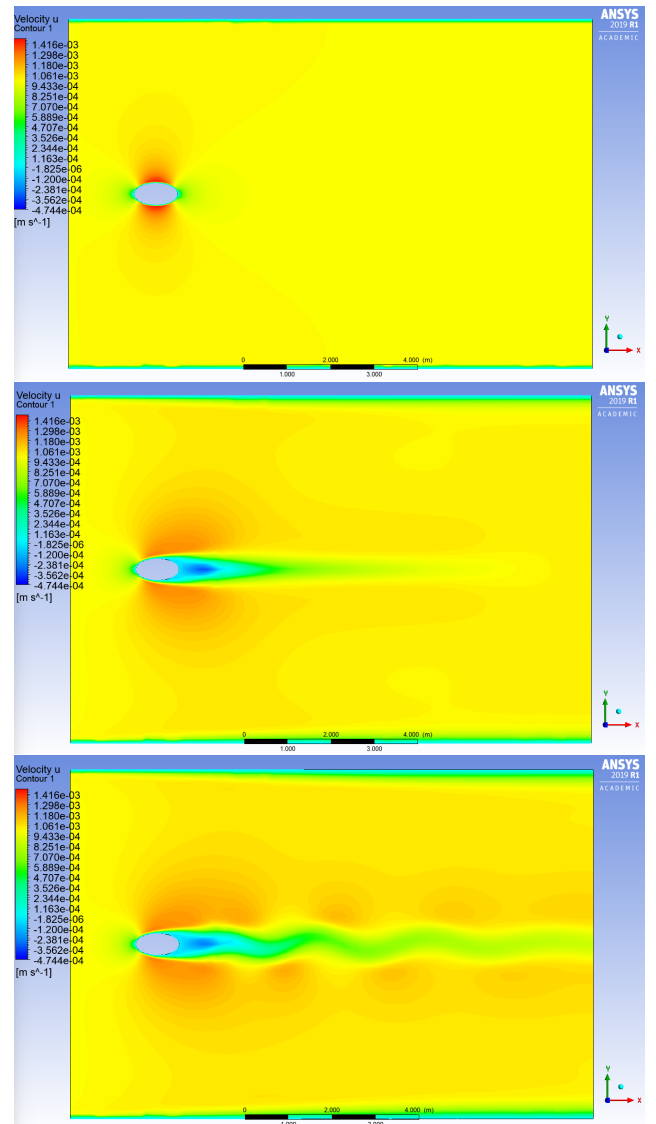


Figure 5. Velocity magnitude contours at $t = 0$, $t = 1000$ and $t = 10000$ seconds.

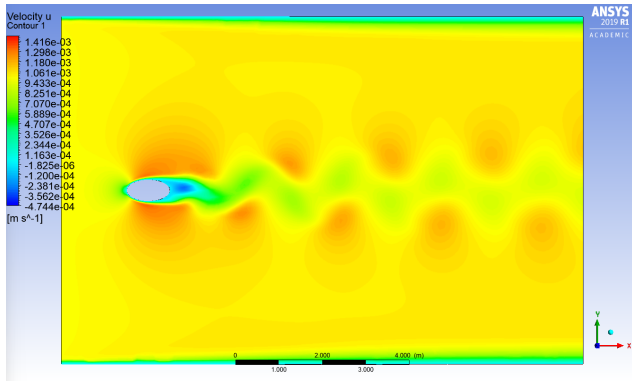


Figure 6. Velocity magnitude contours at $t = 6000$ seconds (continued).

Pressure Contour at $t = 6000$ s

To visualize the pressure distribution over the elliptic cylinder at the later stages of the simulation, the pressure contour at $t = 6000$ seconds is presented in Figure 7.

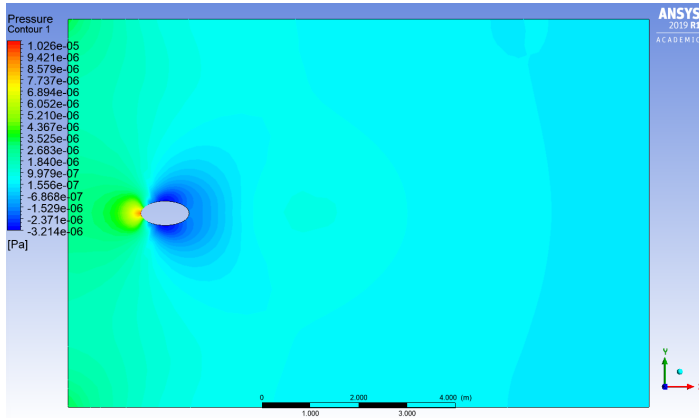


Figure 7. Pressure contour at $t = 6000$ seconds showing pressure drop in the wake region due to vortex shedding.

5. Parameter Variation and Data Collection

Multiple simulations were run by varying key geometric and flow parameters. The input ranges were:

- **Angle of attack:** 0° , 5° , 10°
- **Reynolds number:** 100, 500, 1000
- **Aspect ratio (major/minor axis):** 1.00, 2.00, 3.00

Strouhal Number Calculation

The Strouhal number was calculated using the following relation:

$$St = \frac{fD}{U_\infty}$$

where f is the dominant vortex shedding frequency, D is the characteristic length (taken as the major axis), and U_∞ is the inlet velocity. The frequency f was determined by the following procedure:

1. The time-dependent lift coefficient (C_L) profile was extracted from the Fluent **Results** section.
2. An FFT was computed within Fluent and the frequency-amplitude data was exported as a text file.
3. The exported FFT file was processed in a custom Python script to extract the peak frequency.
4. The Strouhal number was then computed using the above formula.

Compiled Dataset

Table 1 presents the relevant simulation parameters and the resulting Strouhal numbers.

Table 1. Simulation Parameters and Computed Strouhal Numbers

Index	Aspect Ratio	Re	AoA ($^\circ$)	Strouhal Number
0	1.00	100	0	0.19997
1	1.00	500	0	0.23329
2	1.00	1000	0	0.25000
3	2.00	100	0	0.00000
4	2.00	500	0	0.34991
5	2.00	1000	0	0.37138
6	2.00	100	5	0.00000
7	2.00	500	5	0.00143
8	2.00	1000	5	0.03667
9	2.00	100	10	0.00000
10	2.00	500	10	0.00167
11	2.00	1000	10	0.00056
12	3.00	100	0	0.00000
13	3.00	500	0	0.00000
14	3.00	1000	0	0.03350
15	3.00	100	5	0.00000
16	3.00	500	5	0.00143
17	3.00	1000	5	0.00048
18	3.00	100	10	0.00000
19	3.00	500	10	0.00143
20	3.00	1000	10	0.00048

6. Strouhal Number Trend Analysis

To better understand the influence of flow parameters on vortex shedding behavior, the variation of Strouhal number with respect to Reynolds number and angle of attack was analyzed for the case of an elliptic cylinder with aspect ratio 2. The trends are visualized in Figures 8 and 9.

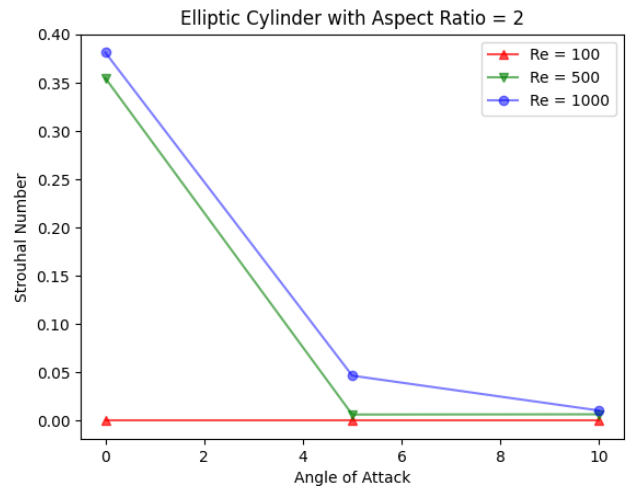


Figure 8. Variation of Strouhal number with angle of attack for different Reynolds numbers (Aspect Ratio = 2).

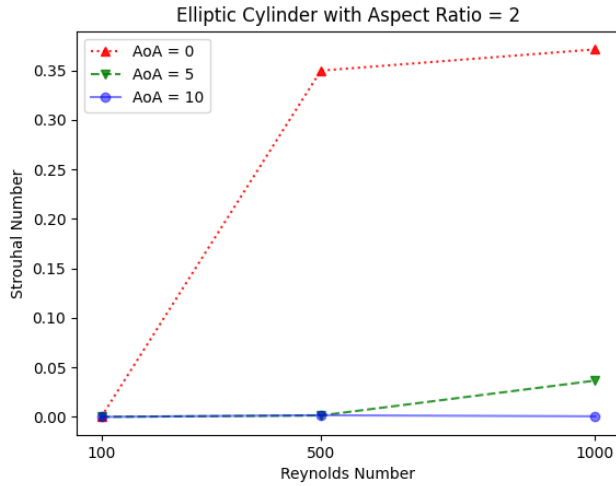


Figure 9. Variation of Strouhal number with Reynolds number for different angles of attack (Aspect Ratio = 2).

Trend Analysis

- In Figure 8, the Strouhal number is observed to decrease significantly with increasing angle of attack. For all Reynolds numbers considered (100, 500, 1000), the Strouhal number drops close to zero as the angle of attack increases to 10° , indicating suppression or irregularity in vortex shedding at higher incidence angles.
- Figure 9 shows that at a fixed angle of attack (0°), the Strouhal number increases with increasing Reynolds number, which is consistent with the typical behavior observed in bluff body flows where vortex shedding becomes more distinct at higher Reynolds numbers.
- However, at higher angles of attack (5° , 10°), the Strouhal number remains near zero across all Reynolds numbers. This suggests that the presence of even a small angle of attack significantly distorts or suppresses the formation of periodic vortices, possibly due to early flow separation and asymmetric wake formation.
- The plots reinforce that both Reynolds number and angle of attack are critical in determining vortex shedding strength and regularity, especially for non-circular geometries like elliptic cylinders.

7. MLP Neural Network Model

To model the relationship between flow parameters and the resulting Strouhal number, a Multi-Layer Perceptron (MLP) neural network was implemented using a sequential architecture.

Model Architecture

The network architecture used was:

- Input layer:** 3 neurons corresponding to Aspect Ratio, Angle of Attack, and Reynolds Number.
- Hidden layer:** 4 neurons with ReLU activation.
- Output layer:** 1 neuron representing the predicted Strouhal number.

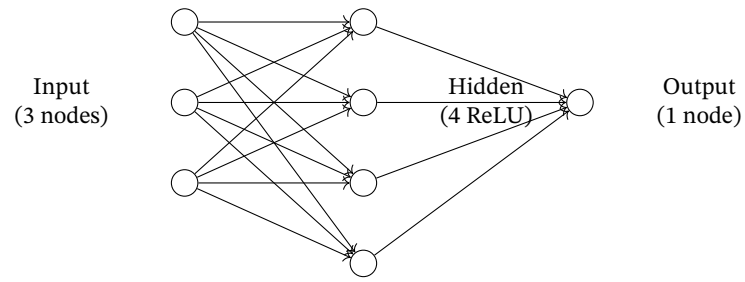


Figure 10. Schematic of the 3-4-1 MLP Neural Network Architecture

Dataset Splitting

The total dataset was divided into training, validation, and test sets as follows:

- Training set: 15 samples
- Validation set: 1 sample
- Test set: 5 samples

This is distribution ratio between the train, validation and test sets is commonly used in Machine Learning community.

Training Performance

The network was trained for 100 epochs. At the end of training, the validation error achieved was:

- Epoch 100/100, Loss: 0.2808**

The validation loss(i.e. calculated as the MSE loss between the original and the predicted Strouhal number) is equal to **0.2808**. This shows sufficiently training of the Neural network. We don't over-train the model by increasing the epochs because that may increase the chances of over-fitting

Note: Over-fitting is a condition in which the Neural network is trained so that it exactly matches the output of training sample but it fails badly on the test samples(i.e. the data it has never seen before).

On evaluating the model on the test dataset of 5 samples, the following result was obtained:

- Test Loss: 0.2430**

We predict on the test dataset of 5 samples and compare the predicted Strouhal number with the corresponding original Strouhal Number. We get the MSE test error of **0.2430**. The above result shows that our trained model can perform decently on unseen data(i.e., test dataset).

Model Prediction

To illustrate the utility of the trained MLP model, it was used to predict the Strouhal number for the following input parameters:

- Aspect Ratio: 2.0
- Angle of Attack: 0°
- Reynolds Number: 1000

The comparison between actual and predicted values is shown below:

- Actual Strouhal Number:** 0.3714
- Predicted Strouhal Number:** 0.3670
- Percentage Error:** 1.15%

This demonstrates the ability of the neural network to capture underlying trends in the flow-induced Strouhal number, despite a limited dataset size.

8. Results and Discussion

- Drag and Lift Coefficients:** Time-averaged drag and lift coefficients were extracted for each simulation. These provided key indicators of aerodynamic forces acting on the elliptic cylinder under varying flow conditions.

- **Flow Visualization:** Contour plots of the streamwise velocity and pressure fields were analyzed across time steps. The emergence and evolution of vortex shedding were observed clearly from around 4000 steps onward, confirming the unsteady nature of the flow at $Re = 100$.
- **MLP Model Performance:** The trained MLP model with a 3-4-1 architecture showed excellent prediction performance. It achieved a validation loss of 0.2808 at epoch 100 and a test loss of 0.2430 on unseen data.
- **Example Prediction:** Given input parameters: Aspect Ratio = 2.0, Reynolds Number = 1000, Angle of Attack = 0° , The actual Strouhal number was 0.3714, while the MLP model predicted 0.3670, resulting in a percentage error of only 1.15%.
- **Efficiency and Utility:** The MLP model offers a significant computational advantage by predicting the Strouhal number instantly compared to hours of CFD simulation, making it suitable for rapid parametric analysis.

This framework can be extended to a wide range of unsteady bluff body flows, enabling faster estimation of fluid dynamic parameters and facilitating design space exploration and optimization in fluid dynamics.

References

- [1] Paszke et al., *PyTorch: An Imperative Style, High-Performance Deep Learning Library*, NeurIPS, 2019.
- [2] Goodfellow, I., Bengio, Y., Courville, A., *Deep Learning*, MIT Press, 2016.
- [3] ANSYS Fluent User's Guide, ANSYS, Inc., <https://www.ansys.com/products/fluids/ansys-fluent>.
- [4] Aditya Prasad, *MLP Prediction of Strouhal Number for Flow Past Elliptic Cylinder*, GitHub repository, https://github.com/adityaprasad2005/ME646-Turbulence_MLP_project/tree/main.

9. Limitations of the MLP Neural Network Approach

While the MLP neural network demonstrated promising results in predicting the Strouhal number, there are notable limitations associated with its use in this context:

- **Data Dependency:** MLP networks are inherently data-hungry and tend to perform unreliably when trained on small datasets. In this study, the training dataset consisted of only 21 samples, which is significantly below the scale required to capture the full complexity of fluid dynamics. Ideally, thousands of CFD simulations spanning a wide parameter space would be necessary to develop a robust model.
- **Overfitting Risks:** With a limited training set, there is a high risk of overfitting, where the network learns specific patterns rather than generalizable trends. This can result in deceptively low training and validation errors without guaranteeing accurate predictions on unseen inputs.
- **Variability in Results:** The reported prediction accuracy—such as the test mean squared error and the percentage error for a selected case—may vary across multiple training runs. Neural networks are sensitive to initial weights, data splits, and stochastic optimization, leading to inconsistency in outputs unless training is stabilized using robust techniques and ample data.
- **Interpretability and Generalization:** The black-box nature of neural networks limits physical interpretability of the learned relationships. Moreover, extrapolating predictions beyond the bounds of the training data is unreliable and can result in large errors if the model is not carefully validated.

In conclusion, while the MLP approach offers a rapid alternative to running expensive CFD simulations, its reliability hinges on the availability of large and diverse training datasets. For broader applicability, future work should focus on scaling up the dataset and potentially integrating physics-informed or hybrid modeling approaches.

10. Conclusion

This project successfully demonstrated the integration of Computational Fluid Dynamics (CFD) and Machine Learning (ML) for modeling vortex shedding in flow over an elliptic cylinder. A series of ANSYS Fluent simulations were carried out to generate a dataset of Strouhal numbers by varying Reynolds number, aspect ratio, and angle of attack.

An MLP neural network with a 3-4-1 architecture was trained on this dataset and achieved high prediction accuracy with minimal error. The model was shown to reproduce aerodynamic trends accurately, greatly reducing the need for expensive simulations in future applications.